



Incident handler's journal

Date: 06/04/2026	Entry: 01
Description	Documenting a cybersecurity incident. This security incident was handled across two distinct operational phases: • Detection and Analysis: The initial phase involved the organisation identifying the ransomware activity on the network. Recognising the complexity of the threat during the analysis stage, internal teams immediately engaged external organisations to secure specialised technical assistance. • Containment, Eradication, and Recovery: Immediate containment protocols were executed, most notably the proactive shutdown of corporate computer systems to isolate the threat. Due to the scale of the compromise, the organisation determined that independent eradication and recovery were not feasible, prompting further escalation to external security partners to assist with full system restoration.
Tool(s) used	None.
The 5 W's	• Who: An organised group of unethical hackers. • What: A ransomware attack. The threat actors distributed phishing emails containing malicious attachments; at least one employee downloaded an attachment, allowing the malware to infect the local device before propagating across the network. • When: Tuesday at 09:00 am. • Where: A small U.S. healthcare clinic. • Why: The incident occurred because unethical hackers successfully breached the organisation's systems via a phishing campaign. Upon gaining access, the attackers deployed ransomware across internal

	systems, encrypting critical files. The attackers' motivation appears to be strictly financial, as the recovery note demanded a large ransom in exchange for the decryption key.
Additional notes	1. A formal assessment of the company's security posture and a "lessons learned" session must follow to establish how the healthcare clinic can prevent an incident of this nature from recurring. This attack could have been prevented through robust security awareness training and effective email vetting procedures. 2. Long-term resilience measures should be implemented, such as establishing robust backup strategies and defining explicit Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs) to minimise operational downtime during future incidents. 3. A critical strategic decision remains regarding whether the company should pay the ransom to retrieve the decryption key, weighing compliance and ethical risks against data recovery needs.

Date: 30/04/2026.	Entry: 02
Description	Documenting a security incident involving a suspicious file hash investigation. An employee received a phishing email containing a password-protected spreadsheet attachment, with the password provided in the body of the email. Upon downloading the file and entering the credential, a malicious payload was

	executed on the local machine. A cryptographic hash of the file was subsequently generated to facilitate a technical investigation.
Tool(s) used	For this activity, I used VirusTotal, which is an investigative tool that analyses files and URLs for malicious content such as viruses, worms, trojans, and more. It's a useful tool for quickly checking whether an indicator of compromise, such as a website or file, has been reported as malicious by others in the cybersecurity community. I used VirusTotal to analyse a file hash, which was reported as malicious. This incident occurred in the Detection and Analysis phase. The scenario put me in the place of a security analyst at a SOC investigating a suspicious file hash. After the suspicious file was detected by the security systems in place, I had to perform deeper analysis and investigation to determine if the alert signified a real threat.
The 5 W's	• Who: An unknown malicious actor. • What: A phishing email containing a password-protected spreadsheet with an embedded malicious executable. The payload generated a SHA-256 file hash of 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b. • When: At 01:20 pm, an alert was generated and sent to the organisation's SOC after the intrusion detection system flagged the file execution. • Where: A financial services company, on an employee computer. • Why: The threat actor targeted the organisation to compromise an internal endpoint and gain unauthorised access to sensitive data. The attack succeeded because the organisation's security controls allowed the malicious email to reach the recipient, enabling the employee to download and execute the attachment.
Additional notes	1. A comprehensive security audit should be conducted at the financial services firm to evaluate existing perimeter boundaries and endpoint defence measures.

	2. Advanced Threat Protection (ATP) solutions must be explored, such as Secure Email Gateways (SEGs), API-based cloud email security, or Next-Generation Firewalls (NGFWs). Implementing these controls will establish a layered, defence-in-depth posture capable of inspecting encrypted attachments or flagging anomalous inbound sender behaviour. 3. Targeted security awareness training should be formalised for all personnel, specifically focusing on identifying sophisticated social engineering tactics, such as password-protected attachments, and reinforcing internal reporting protocols for suspicious emails.
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Date: 07/05/2026	Entry: 03
Description	Analysing a packet capture file
Tool(s) used	For this activity, Wireshark was utilised to analyse a packet capture file. Wireshark is a network protocol analyser with a graphical interface, valued in cybersecurity for its ability to capture and inspect network traffic. By allowing security analysts to examine live or recorded data packets, the tool serves as a critical asset for detecting and investigating malicious activity across a network.
The 5 W's	Capture the 5 W's of an incident. • Who: N/A • What: N/A • When: N/A • Where: N/A • Why: N/A
Additional notes	This was my first time using Wireshark, so I was excited to complete this exercise and analyse a packet capture file. At first glance, the interface was

	challenging, but I enjoyed learning how to use it and getting hands-on experience with Wireshark. I can see why it's such a powerful tool for understanding network traffic.
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Date: 15/05/2026	Entry: 04
Description	Capturing my first packet
Tool(s) used	For this activity, tcpdump was utilised to capture and analyse network traffic. Accessible via the command-line interface, tcpdump is a network protocol analyser that, similar to Wireshark, provides essential value in cybersecurity by allowing security analysts to capture, filter, and inspect network traffic data.
The 5 W's	• Who: N/A • What: N/A • When: N/A • Where: N/A • Why: N/A
Additional notes	I'm still new to using the command-line interface, so using it to capture and filter network traffic was a challenge. I got stuck a couple of times because I used the wrong commands. However, after revisiting the learning materials and repeating some of the steps, I was able to complete the activity and successfully capture network traffic.

Date: 20/05/2026	Entry: 05
Description	Documenting a data breach incident and subsequent technical investigation. A web application vulnerability enabled an external threat actor to bypass authorisation controls and gain unauthorised access to customer Personally Identifiable Information (PII) and financial records, impacting approximately 50,000 customer records. The financial impact is estimated at \$100,000 in direct costs and potential revenue loss. The root cause was isolated to the e-commerce platform's URL parameter handling, allowing the attacker to systematically scrape sensitive transaction data.
Tool(s) used	SIEM Tool & Wireshark: Utilised within the Detection and Analysis phase to review web server access logs and network traffic data, which confirmed a single log source generating an exceptionally high volume of sequentially listed customer order requests.
The 5 W's	Capture the 5 W's of an incident. • Who: An independent unethical hacker. • What: An employee received an email stating that they had stolen customer data, demanding a ransom. The root cause of the incident was identified as a vulnerability in the e-commerce web application. This vulnerability allowed the attacker to perform a forced browsing attack and access customer transaction data by modifying the order number included in the URL string of a purchase confirmation page. This vulnerability allowed the attacker to access customer purchase confirmation pages, exposing customer data, which the attacker then collected and exfiltrated. The logs indicated that the attacker accessed the information of thousands of purchase confirmation pages. • When: An initial ransom email was received at 03:13 pm on 22 December 2022 but was dismissed as spam. • A second email containing a data sample was received on 28 December 2022, prompting an investigation. The formal security incident and

	unauthorised access footprint were established on 28 December 2022 at 19:20 PT. • The investigation concluded on 31 December 2022. • Where: A mid-sized retail company's public-facing e-commerce web application. • Why: The attack was motivated by financial gain, as evidenced by an initial \$25,000 cryptocurrency demand that escalated to \$50,000 upon the second contact.
Additional notes	1. Defensive Gaps & Detection: This incident highlights a critical gap in behavioural alerting. A single external source was permitted to query thousands of sensitive customer pages sequentially without triggering a rate-limiting threshold or generating a high-priority SOC alert. A defence-in-depth approach is needed; a SIEM system and an IPS tool working in tandem could have flagged this through behavioural analysis, allowing the IPS to dynamically block the IP address and limit the damage. 2. Remediation & Response Actions: Following containment, the organisation collaborated with public relations to disclose the data breach transparently to customers. To mitigate external impact and preserve brand reputation, free identity protection services were offered to all affected individuals. 3. Strategic Resilience: The immediate recommendations proposed, including routine vulnerability scanning, penetration testing, strict URL allowlisting, and enforcing robust authentication requirements, are critical to hardening the perimeter. However, current security systems must be continuously revised to ensure that if these initial authorisation controls are ever bypassed again, active monitoring and automated response capabilities are in place to rapidly detect and contain threats, helping to minimise their impact and strengthen the organisation's overall security posture.

Reflections/Notes:

1. Technical Progression and Command-Line Adaptation

While expanding my hands-on experience with command-line utilities, configuring tcpdump using the correct command syntax and filtering flags required a methodical approach. Rather than relying on graphical interfaces, adapting to the CLI environment required a structured approach to accurately configure packet capture parameters. By systematically auditing my syntax against expected network behaviours and breaking down the commands into modular components, I quickly achieved a solid working understanding. This process reinforced a critical engineering principle: accurate packet capture and log collection depend on precise command syntax to produce reliable results.

2. Operational Evolution of Incident Response

My understanding of the incident response lifecycle has progressed from a conceptual foundation to a more structured, practical methodology. Developing these journals reinforced that effective incident response depends on structured processes, clear documentation, and the appropriate use of security tools, rather than isolated technical actions. Navigating the phases from initial detection to long-term eradication highlighted how an organisation's security posture relies on well-defined incident response playbooks, clear communication procedures, and defensive frameworks to limit the spread and operational impact of an incident, reducing potential damage to the organisation.

3. Telemetry Analysis and Risk Mitigation

Exploring network traffic analysis and log auditing proved to be one of the most engaging aspects of this technical curriculum. Gaining exposure to protocol analysers provided valuable insight into network traffic, infrastructure communications, and technical troubleshooting, and how technical evidence can be used to identify potential security risks. Developing this foundation has solidified my focus on defensive security operations, technical vendor support, and risk governance. I am actively expanding my skill set to bridge the gap between technical system diagnostics and the broader controls required to maintain an organisation's overall security and compliance posture.